

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (E&E) (Credit System) Degree Examinations**

January 2023

20EE504 – POWER ELECTRONICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.***Unit – I**

	Marks	BT*	CO*	PO*
1. a) Discuss the various peripheral effects of power electronic converters.	6	L*2	1	1,2
b) Draw and explain the switching characteristics of power BJT.	6	L1	1	1,2
c) Discuss the need for isolation of gate and base drive. Also discuss how pulse transformer can be used for isolating the gate signals	8	L2	1	1,2
2. a) Draw the control characteristics of GTO, SCR, and BJT.	6	L1	1	1,2
b) The bipolar transistor switching circuit with β in the range of 10 to 40. If $R_C=12\Omega$, $V_{CC}=220V$, $V_B=12V$, $V_{CE(sat)}=V_{BE(sat)}=1.2V$, find (i) the value of R_B to move the transistor in to saturation with an ODF of 4, (ii) The forced β of the transistor, and (iii) The power loss PT in the transistor.	6	L3	1	1,2
c) What is di/dt and dv/dt protection? Discuss the methods for di/dt and dv/dt protection of a transistor.	8	L2	1	1,2

Unit – II

3. a) Discuss the characteristics of UJT and its operation as relaxation oscillator	6	L2	2	1,2
b) Ten thyristors are used in a string to withstand a DC voltage of $V_S=15kV$. The maximum leakage current and recovery charges of thyristors are 12mA and 160 μC respectively. Each thyristor has a voltage sharing resistance of $R=60k\Omega$ and capacitance $C_1=1\mu F$. Find i) Maximum steady state voltage sharing. ii) Steady state voltage derating factor (DRF). iii) Maximum transient voltage sharing. iv) Transient voltage derating factor	6	L3	2	1,2
c) With the help of a neat circuit diagram and waveforms explain the working of complementary commutation circuit to commutate a thyristor. Also derive an expression for turn off time.	8	L2	2	1,2
4. a) Define latching and holding current. Also draw the i-v characteristics of SCR	6	L1	2	1,2
b) Discuss the two transistor model of an SCR and hence deduce the expression for its anode current.	6	L2	2	1,2
c) What is commutation? With neat circuit and waveforms discuss the operation of self-commutation of thyristor.	8	L2	2	1,2

Unit – III

5. a) With a neat circuit diagram and waveforms discuss the operation of single phase full wave controller with inductive load. Also derive an expression for rms value of output voltage.	10	L2	3	1,2,3,4,
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P.T.O.

- b) A step down chopper with resistive load has load resistance of 10Ω , input voltage $V_s=220V$. Assuming a duty cycle of 50%, on state voltage drop of 2V and switching frequency of 1kHz. Estimate
- Rms value of output voltage and current
 - Average value of output voltage and current
 - Chopper efficiency
- 10 L3 3 1,2,3,4,
6. a) Discuss the operation of step-up chopper with RL load. Also derive an expression for peak-to-peak ripple current
- 10 L2 3 1,2,3,4,
- b) A single-phase full wave ac voltage controller has a resistive load of 25Ω and input voltage is 220V, 50Hz. The delay angle of both the thyristors is $\pi/2$ (90°). Determine
- Rms output voltage and current
 - Input power factor
 - The average current of thyristor
 - RMS current of thyristor
- 10 L3 3 1,2,3,4,

Unit – IV

7. a) With a neat circuit and waveforms discuss and analyse the principle of three phase half wave converter with pure resistive load. Also derive an expression for average output voltage
- 10 L2 4 1,2,3,4,12
- b) If a single-phase controlled converter has a pure resistive load of $R\Omega$ and delay angle of $\pi/2$ (90°). Determine
- Rectification efficiency
 - Form factor
 - Ripple factor
 - Transformer utilization factor
- 10 L3 4 1,2,3,4,12
8. a) Discuss the working of single phase two quadrant converter feeding highly inductive load. Sketch the input, output, and the source current waveform assuming a constant, continuous, ripple free load current. Derive an expression for RMS value of output voltage
- 10 L2 4 1,2,3,4,12
- b) A three phase full converter operated from a 230V, 50Hz supply with a resistive load of 20Ω . It is required to obtain 50% of maximum possible output voltage. Calculate
- Delay angle
 - RMS and average currents
 - Power factor
 - Rectification efficiency
- 10 L3 4 1,2,3,4,12

Unit – V

9. a) Draw the circuit diagram of a three phase bridge inverter with Y connected resistive load. Sketch the gating signals and line to line output voltage for 180° conduction operation.
- 10 L2 5 1,2,3,12
- b) Explain the following with reference to voltage control of single-phase inverter. (i) Multiple PWM, and (iii) Sinusoidal PWM
- 10 L2 5 1,2,3,12
10. a) With a neat circuit diagram explain the working of single-phase full-bridge inverter supplying to resistive load.
- 7 L2 5 1,2,3,12
- b) Discuss the various performance parameters of an inverter.
- 6 L2 5 1,2,3,12
- c) Discuss the operation of variable DC link inverter.
- 7 L2 5 1,2,3,12

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
